on the Earth-based "geocentric" universe, but the perfect-circle part turned out to be a bit off—all planets orbit the Sun in flattened circles called ellipses, and even that shape is just an approximation of a more complex trajectory. Copernicus's basic idea was correct, and that's what mattered most. It simply required some tweaking to make it more accurate.

Yet, in the case of Einstein's relativity, the founding principles of the entire theory require that everything must happen exactly as predicted. Einstein had, in effect, built what looks on the outside like a house of cards, with only two or three simple postulates holding up the entire structure. Indeed, upon learning of a 1931 book entitled *One Hundred Authors Against Einstein*,[††] he responded that if he were wrong, then only one would have been enough.

Therein were sown the seeds of one of the most fascinating blunders in the history of science. Einstein's new equations of gravity included a term he called the "cosmological constant," which he represented by the capital Greek letter lambda: Λ. A mathematically permitted but optional term, the cosmological constant allowed him to represent a static universe.

Back then, the idea that our universe would be doing anything at all, other than simply existing, was beyond anyone's imagination. So lambda's sole job was to oppose gravity within Einstein's model, keeping the universe in balance, resisting the natural tendency for gravity to pull the whole universe into one giant mass. In this way, Einstein invented a universe that neither expands nor contracts, consistent with everybody's expectations at the time.

The Russian physicist Alexander Friedmann would subsequently show mathematically that Einstein's universe, though balanced, was in an unstable state. Like a ball resting on the top of a hill, awaiting the slightest provocation to roll down in one direction or another, or like a pencil balanced on its sharpened point, Einstein's universe was precariously perched between a state of expansion and total collapse. Moreover, Einstein's theory was new, and just because you give something a name does not make it real—Einstein knew that lambda, as a negative gravity force of nature, had no known counterpart in the physical universe.

✶